Independent Program Verification Outperforms Translation in Diagnosing Reasoning Errors in Small Language Models: A Causal and Metacognitive Framework

Huynh Nhat Huy
Independent Researcher
cathuys69@gmail.com

November 14, 2026

Abstract

Chain-of-thought (CoT) prompting frequently suffers from post-hoc rationalization, wherein intermediate verbalized rationales fail to causally govern model answers or merely mimic flawed heuristic assumptions. In this paper, we investigate whether deterministic programmatic verifiers and structured reasoning representations can reliably diagnose reasoning fallacies and execution slips in fine-tuned 8-billion parameter models (Qwen3-8B 4-bit with LoRA) without human intervention. We establish a pre-registered experimental evaluation across 500 challenging problems from the MATH dataset (Levels 3–5) and GSM8K. On GSM8K, structured `Step/Calc` fine-tuning achieves high zero-prefill format validity (88/100–96/100 across three seeds of `enriched_cf`; one seed misses the 90/100 gate) and high adherence to intermediate arithmetic perturbations (89.7%–93.8% following, with 2.8%–3.1% reverting). Accuracy, however, stays below `base_prefill` (93/100): three seeds of `enriched_cf` reached 87, 82 and 82/100 accuracy, and two of the three are significantly lower (McNemar $p = 0.0034$ and 0.0074; $p = 0.1094$ for the best seed). On the sealed GSM8K test (1,319 questions, run once under a protocol registered before the test file was downloaded), counterfactual training makes the answer follow an edited intermediate result more often than plain SFT (544/671 vs. 512/671, exact two-sided McNemar $p = 0.0018$) with no detectable accuracy difference (79.8% vs. 78.8%, $p = 0.46$), but both stay well below `base_prefill` (88.5%), so the pre-registered non-inferiority criterion fails. These results replicate known effects of counterfactual and plain fine-tuning; in a post-hoc comparison, plain SFT follows edits less often than the base model while counterfactual training stays close to it. On symbolic mathematics, flat structured traces run into problems: internal equational verification is insufficient, catching only 2.4% (2/82) of erroneous solutions on MATH Levels 3–5 (items 101–300 under `base_showcalc`, false alarm rate 0/118), and although non-truncated LoRA traces achieve 72.7% (40/55) accuracy, 75.6% (34/45) of truncated sequences end in greedy repetition loops; removing one-step training traces did not reduce truncation in a controlled ablation (20 vs. 25 of 100, $p = 0.42$). In one pre-registered development run on MATH (items 301–500; two automated labelling passes over 39 solutions, $\kappa = 0.947$, no human review), the results are consistent with the **Circular Confirmation Trap**: **independent programmatic generation catches 5× more reasoning errors than literal thought-trace translation** (10/24 [41.7%] vs. 2/24 [8.3%], exact two-sided McNemar $p = 0.0215 < 0.05$), while translation achieves parity on pure execution arithmetic (69.2% vs. 61.5%). To resolve the flat-trace repetition versus monolithic-interpreter crash trade-off on symbolic mathematics, we outline an encapsulated Directed Acyclic Graph (DAG) and selective tool-delegation architecture (MSRA) as future work.

1 Introduction

Modern autoregressive large language models demonstrate competitive performance when generating natural language rationales step by step [Wei et al., 2022, Cobbe et al., 2021]. Despite these empirical

gains, extensive evidence reveals that verbalized rationales often function as post-hoc justifications rather than faithful computational engines [Lanham et al., 2023, Turpin et al., 2024]. In complex mathematical problem solving, reasoning errors fall into two fundamentally distinct categories:

1. **Execution Slips:** Arithmetic miscalculations where the underlying logical formula or conceptual relationship is correct, but the discrete evaluation is erroneous (e.g., $14 \times 7 = 96$).

2. **Reasoning Fallacies:** Conceptual formulation breakdowns where the model asserts an invalid mathematical premise, ignores physical constraints, or constructs a fundamentally incorrect equation (e.g., equating perimeter directly to area).

To mitigate arithmetic failure modes, prior works such as Program-Aided Language Models (PAL) [Gao et al., 2023] and Program of Thoughts (PoT) [Chen et al., 2023] outsource arithmetic evaluation to an external Python interpreter. When applying programmatic verification to monitor autoregressive generation, two divergent paradigms emerge:

- **Translated Trace Verification ($\mathcal{T}$):** The model translates its natural language thought trace into Python/SymPy code, which is then executed to verify whether the textually claimed answer matches the executed value.

- **Independent Program Verification ($\mathcal{I}$):** An external solver script is generated directly and independently from the problem text Q, bypassing the internal textual rationale.

A central dilemma arises: *Does translating the model's textual thinking into code enhance diagnostic sensitivity, or does it merely automate the transcription of the model's hallucinations?*

In this work, we address these fundamental questions through a multi-phase pre-registered study on fine-tuned Qwen3-8B [Yang et al., 2024a] across multi-step arithmetic (GSM8K) and challenging symbolic mathematics (MATH Levels 3–5) [Hendrycks et al., 2021]. We establish three core findings:

1. **Structured Traces & Causal Faithfulness on Arithmetic:** Fine-tuning with explicit `Step/Calc` delimiters yields high format compliance (88/100–96/100 over three seeds of `enriched_cf`, one seed below the Gate 0 threshold) and high causal faithfulness under intermediate arithmetic perturbations (89.7%–93.8% following rate, with only 2.8%–3.1% reverting to prior unperturbed answers). Combining Step 1 semantic enrichment with counterfactual augmentation (`enriched_cf`) does not show an accuracy gain over the unenriched adapter: three seeds of `enriched_cf` reached 87, 82 and 82/100 accuracy (mean 83.67), against 83/100 for `old_cf` and 93/100 for `base_prefill`. On the sealed GSM8K test, `enriched_cf` follows an edited step more often than `enriched_plain` (81.1% vs. 76.3% of 671 shared episodes, $p = 0.0018$), at equal accuracy and 8.7 points below `base_prefill` (Section 3.4). This replicates, under a pre-registered protocol, effects reported by Swaroop et al. [2025] and Lobo et al. [2024]; a post-hoc comparison suggests counterfactual training keeps step-following near the base model's level rather than raising it.

2. **The Flat-Trace Bottleneck on Symbolic Mathematics:** When scaling flat `Step/Calc` traces to MATH Levels 3–5, simple internal equational verification fails, catching only 2.4% (2/82) of erroneous solutions. Moreover, while non-truncated LoRA completions achieve 72.7% (40/55) accuracy, 75.6% (34/45) of truncated generations end in greedy repetition loops. We have not identified the cause: a controlled ablation that removed one-step training traces did not reduce truncation (Section 3.5).

3. **The Circular Confirmation Trap:** We state a simple bound (Theorem 1) for why high-fidelity translation tends to copy a flawed premise, and report a consistent pattern on one pre-registered development run: translated checking caught only 8.3% of reasoning bugs (two automated labelling passes over 39 solutions, $\kappa = 0.947$, no human review). In contrast, independent programmatic verification exposes reasoning fallacies with 5× higher recall

(41.7%, exact two-sided McNemar $p = 0.0215$), while matching translation on execution slips (61.5% vs. 69.2%).

Finally, to address the dilemma where flat autoregressive traces collapse into repetition loops while monolithic programmatic solvers fail execution on complex symbolic queries (27/100 crashes in PAL), we outline an encapsulated Directed Acyclic Graph (DAG) reasoning agent with reference-free preference alignment (MSRA) as a principled direction for future work.

2 Related Work & Theoretical Foundations

2.1 Rationale Faithfulness and Post-Hoc Confabulation

The causal relationship between language model rationales and downstream answers has been extensively questioned. Lanham et al. [2023] demonstrated via feature attribution and prompt perturbation that models frequently generate identical answers even when extensive portions of their reasoning traces are truncated or modified. Turpin et al. [2024] revealed that injecting biased features into the context causes models to systematically alter their answers while systematically omitting the biasing influence from their stated rationales. Lyu et al. [2023] proposed restructuring rationales into deterministic dependency chains, though primarily evaluated on synthetic tasks without assessing error diagnosis.

2.2 Training for Faithful Reasoning

Several works fine-tune models so that answers depend on the stated steps. FRODO [Paul et al., 2024] trains a reasoner with a counterfactual preference objective and reports more faithful rationales than supervised fine-tuning (SFT). LIFT [Swaroop et al., 2025] replaces a step with an irrelevant fact, continues the chain, and fine-tunes LoRA adapters on traces whose answer changed; it uses Qwen3-8B and GSM8K, the same model and benchmark as our study, and compares against an SFT baseline. Causal consistency regularization [Akter et al., 2025] builds counterfactual traces by editing operators (for example, swapping + for −). Counterfactual simulation training [Hase and Potts, 2026] and CASE [Wang et al., 2026] are more recent variants; CASE also reports that several faithfulness-training methods do not exceed the untuned base model under domain transfer. Lobo et al. [2024] show that ordinary QLoRA SFT, including on GSM8K, lowers CoT faithfulness relative to the base model. Our counterfactual training and our plain-SFT comparison therefore replicate known effects; what we add is a test protocol registered before a sealed test set was opened, an exact-match criterion for following an edited value, and an outcome taxonomy.

2.3 Programmatic Reasoning and Process Verifiers

Outsourcing intermediate computation to deterministic interpreters represents a prominent paradigm for eliminating arithmetic drift. PAL [Gao et al., 2023] and PoT [Chen et al., 2023] demonstrated that generating Python scripts outperforms standard CoT on arithmetic benchmarks. In parallel, Process Reward Models (PRMs) [Lightman et al., 2023, Uesato et al., 2022] train learned neural verifiers to assign step-by-step correctness scores. However, PRMs require millions of step-level human or synthetic annotations and demand substantial inference-time compute. In contrast, our study focuses on lightweight, deterministic programmatic checking that executes in milliseconds without auxiliary neural verifiers.

2.4 Fine-Tuning Small Language Models for Mathematical Reasoning

Recent research has increasingly focused on instilling multi-step mathematical reasoning into compact language models (< 10B parameters), moving away from the assumption that complex problem solving is strictly an emergent capability of massive ≥ 100B models. As summarized in Table 1, existing methodologies span four major paradigms:

1. **Synthetic Data Augmentation and Self-Evolution:** Li et al. [2023] introduced TinyGSM, proving that a 1.3B model fine-tuned on 12.3M synthetic Python-annotated problems can exceed 81.5% accuracy on GSM8K when paired with a verifier. Similarly, MetaMath [Yu et al., 2024] bootstrapped 395K problems via multi-perspective question rewriting and backward reasoning to lift LLaMA-2-7B to 66.4%, while Orca-Math [Mitra et al., 2024] leveraged a multi-agent generation pipeline (200K problems) and iterative preference learning (KTO) to reach 86.8% pass@1 on Mistral-7B without verifiers or execution tools.

2. **Step-by-Step Rationale Distillation:** Rather than relying on standard label fine-tuning, Hsieh et al. [2023] demonstrated that training sub-billion parameter models (e.g., T5-770M) to jointly predict teacher-generated intermediate rationales and task labels drastically improves sample efficiency across reasoning tasks. At a larger scale, DeepSeek-R1 [DeepSeek-AI et al., 2025] demonstrated that distilling 800K long-chain reasoning trajectories generated by a large reinforcement-learned reasoning model into dense student models yields exceptional accuracy: 83.9% on Qwen-1.5B, 92.8% on Qwen-7B, and 89.1% on LLaMA-8B on GSM8K.

3. **Hybrid Tool-Use and Code Synthesis:** Recognizing that pure natural language autoregression struggles with arithmetic and symbolic precision, MAmmoTH [Yue et al., 2024] pioneered hybrid instruction tuning combining Chain-of-Thought (CoT) and Program-of-Thought (PoT) trajectories across diverse math datasets. ToRA [Gou et al., 2024] trained 7B models on interactive tool-use trajectories to seamlessly alternate between reasoning and SymPy/Python computation, reaching 68.8%–72.6% on GSM8K and 44.6% on MATH. Concurrently, MathCoder [Wang et al., 2024a] demonstrated that fine-tuning open-source models to generate code, capture runtime execution results, and continue deriving equations yields 67.8% accuracy on GSM8K.

4. **Automated Process Supervision and Group Relative RL:** Moving beyond sparse outcome rewards, WizardMath [Luo et al., 2025] applied Reinforced Evol-Instruct (RLEIF) with Process Supervised Reward Models (PRMs) to enhance LLaMA-2-7B and Mistral-7B up to 83.2% on GSM8K. Math-Shepherd [Wang et al., 2024b] constructed automated step-wise process verification via Monte Carlo rollouts without human annotation, elevating Mistral-7B to 84.1% via step-wise PPO and 89.1% under verification. Building on rule-based outcome supervision, DeepSeekMath [Shao et al., 2024] introduced Group Relative Policy Optimization (GRPO) to train 7B models to 88.2% on GSM8K, which was subsequently integrated with self-improving reward modeling and tool-integrated reasoning in Qwen2.5-Math [Yang et al., 2024b] to push 1.5B and 7B models to 84.8%–95.2% on GSM8K.

The Structural Gap in Existing SLM Paradigms. Despite these impressive benchmark scores, existing post-training and tool-integrated methodologies almost universally treat accuracy on fixed benchmarks (pass@1 on GSM8K/MATH) as the sole metric of success. Crucially, they leave two fundamental scientific questions unanswered:

1. **Causal Authenticity vs. Memorized Confabulation:** These accuracy-oriented works do not measure whether the downstream answer causally depends on the intermediate steps. That question is addressed by the faithfulness-training work in Section 2.2; our contribution there is a pre-registered, sealed-test replication rather than a new method.

2. **Diagnostic Blindness and the Circular Confirmation Trap:** Prior programmatic and verifier systems assume that either generating code or translating a natural language trace will faithfully catch errors. However, as we formalize in Section 2.5, translating a model's flawed textual thoughts into an executable script faithfully re-encodes the hallucinated premise with probability at least ϕ, rendering translated checking blind to conceptual errors. Our work addresses this blind spot by proposing independent programmatic verification, which achieves 5× higher sensitivity in diagnosing reasoning fallacies while matching translation on arithmetic execution slips.

2.5 Theoretical Analysis: The Circular Confirmation Trap

We provide a formal mathematical model explaining why translated verification fails to catch reasoning bugs.

Definition 1 (Reasoning and Program Variables)**.** *Let $Q \in \mathcal{X}$ denote the input query. Let $T \in \mathcal{T}$ denote the autoregressive natural language thought trace generated by policy $\pi_\theta(T \mid Q)$, and $A \in \mathcal{Y}$ denote the final answer. Let $C_T \sim \pi_{trans}(C \mid Q, T)$ denote the translated program, and $C_I \sim \pi_{indep}(C \mid Q)$ denote the independently generated program. Let $y^* \in \mathcal{Y}$ denote the ground-truth solution.*

Definition 2 (Error Decomposition)**.** *We partition failures $A \neq y^*$ into two disjoint error sets:*

1. *An **Execution Slip** (E_{exec}): The mathematical relations in T are conceptually valid, but manual computation produces an arithmetic error.*
2. *A **Reasoning Fallacy** (E_{reason}): The mathematical relations asserted in T are conceptually false, contradictory, or structurally invalid.*

Theorem 1 (Circular Confirmation Bound in Translation)**.** *Assume policy π_{trans} has semantic fidelity $\phi \in (0, 1)$ such that for any premise $p \in T$, the probability that C_T encodes p is at least ϕ. Then for any reasoning fallacy $E_{reason} \in T$ governed by flawed premise p_{flaw}, the probability that translated execution C_T detects the fallacy satisfies:*

$$\mathbb{P}(\mathit{Detect}(C_T) \mid E_{reason}) \leq 1 - \phi \tag{1}$$

Proof. Let $\mathrm{Eval}(C)$ denote the executed output of program C. By definition, detection occurs when $\mathrm{Eval}(C_T) \neq A$. Under error condition E_{reason}, the model asserts answer $A = f(p_{\text{flaw}})$. If the translator faithfully transcribes p_{flaw} with probability ϕ, the program executes $\mathrm{Eval}(C_T) = f(p_{\text{flaw}}) = A$. Hence, $\mathbb{P}(\mathrm{Eval}(C_T) = A \mid E_{\text{reason}}) \geq \phi$, which implies $\mathbb{P}(\mathrm{Detect}(C_T) \mid E_{\text{reason}}) \leq 1 - \phi$. □

Corollary 2 (Fidelity-Detection Trade-off)**.** *As the language model's code translation capability improves ($\phi \to 1$), its capacity to detect reasoning errors via translated trace checking approaches zero ($\mathbb{P}(\mathit{Detect}(C_T) \mid E_{reason}) \to 0$).*

In contrast, independent programmatic generation C_I is sampled from $\pi_{\text{indep}}(C \mid Q)$, conditionally independent of the natural language trace T:

$$\mathbb{P}(\mathrm{Detect}(C_I) \mid E_{\text{reason}}) = \mathbb{P}(\mathrm{Eval}(C_I) \neq A \mid E_{\text{reason}}) \tag{2}$$

Because the generation of C_I requires constructing an algorithmic solver from scratch, it is not bound to repeat the hallucinated intermediate premise p_{flaw}, achieving strictly higher recall on reasoning bugs.

3 Empirical Evaluation & Pre-Registered Results

3.1 Experimental Setup & Pre-Registration Protocol

All experiments are conducted on Qwen3-8B 4-bit fine-tuned with LoRA ($r = 8$, scale 20.0, dropout 0.0, 8 target layers, learning rate 5×10^{-5}, training `--max-seq-length 512` on 16 GB Apple M1) using MLX.

- **MATH Dataset Partitions:** 1,301 deterministic checkable problems from Levels 3–5 partitioned via SHA-256 hash ordering into 725 train, 76 valid, and 500 dev problems.
- **Strict Disjointness:** The official GSM8K test set was opened once, under a protocol registered beforehand (Section 3.4); none of its questions appear in the training, validation or development splits. MATH-500 remains sealed.
- **Adjudication:** The error taxonomy on MATH items 301–500 comes from two automated labelling passes ($\kappa = 0.947$ between them) with no human review.

3.2 Testing the Three Hypotheses

Hypothesis 1 (Equational Identity Checking): *Simple equational consistency checking catches mathematical errors.* **Outcome: Falsified.** On 200 MATH Level 3–5 dev questions (items 101–300, `base_showcalc`, comprising 118 correct and 82 erroneous solutions), equational consistency checking (Signal A) detected only 2 out of 82 erroneous solutions (2.4%, false alarms 0/118). Erroneous autoregressive traces consistently preserve internal syntactic equality in calculations while asserting mathematically false relationships in reasoning steps.

Hypothesis 2 (Unconstrained Program Verification): *Programmatic checking catches reasoning errors independent of trace semantics.* **Outcome: Rejected on 2026-11-04.**

Hypothesis 3 (Independent vs. Translated Verification on MATH Levels 3–5): *Independent program generation catches reasoning errors that translated code duplicates, whereas translated code catches execution slips at least as effectively as independent programs.*

Outcome: All 5 Pre-Registered Criteria Met on This Development Run ($p = 0.0215$); Not Yet Confirmed on the Sealed Test. Evaluated on 200 previously unread development questions (MATH dev items 301–500). Errors were labelled by two automated passes, made before the checker programs were generated, over 39 solutions ($\kappa = 0.947$). One pass was rerun because its first attempt had seen the other's labels, and no human has reviewed them:

1. **Sample Size Condition:** 24 reasoning errors identified (≥ 20 required).
2. **Reasoning Error Detection:** Independent programmatic generation caught **10/24 (41.7%)**, whereas translated trace checking caught only **2/24 (8.3%)** (Figure 1(a)). McNemar exact two-sided test: $p = 0.0215 < 0.05$.
3. **False Positive Margin:** The gap in reasoning detection (0.334) strictly exceeded the gap in false alarms on correct solutions (0.057, 10/141 [7.1%] vs. 2/141 [1.4%]).
4. **Execution Error Detection:** On 13 execution slips, translated checking caught **9/13 (69.2%)**, while independent checking caught **8/13 (61.5%)**, satisfying the condition of parity ($\geq$).
5. **Composite Ensemble Gate:** Raising an alert when either checker disagreed captured **32/59 (54.2%)** of all failures (95% Wilson CI: 41.7–66.3%) with only 12/141 (8.5%) false alarms on correct answers (Figure 1(b)).

3.3 Causal Faithfulness Interventions & Semantic Enrichment

To determine whether intermediate structured calculations causally govern model conclusions, we execute automated counterfactual interventions by perturbing intermediate `Calc` results across the 100 GSM8K development questions, which were also read during error analysis:

- **Baseline vs. Plain SFT vs. Counterfactual Training:** Standard LoRA fine-tuning (`old_plain`, `gsm8k_plain_2000_s0`) achieved 87/100 accuracy on Dev-100, but only 87/100 format validity (86/100 in strict analyzer, 95% Wilson CI: 77.9–91.5%), failing Gate 0 ($\geq 90\%$). Introducing 3,119 counterfactual training instances (`old_cf`, `gsm8k_cf_2000_s0`) raised format validity to 95/100 (95% Wilson CI: 88.8–97.8%, passing Gate 0), while accuracy dropped to 83/100 (vs. 93/100 for `base_prefill`, McNemar $p = 0.0129$).
- **Semantic Enrichment Ablation (`enriched_plain` & `enriched_cf`):** Error diagnostics on legacy adapters revealed that intermediate breakdowns frequently stemmed from ungrounded Step 1 openings. Enriching the training corpus so that 100% of Step 1 reasoning traces open with explicit variable declarations (`Calculate the <quantity>...`) boosted plain SFT

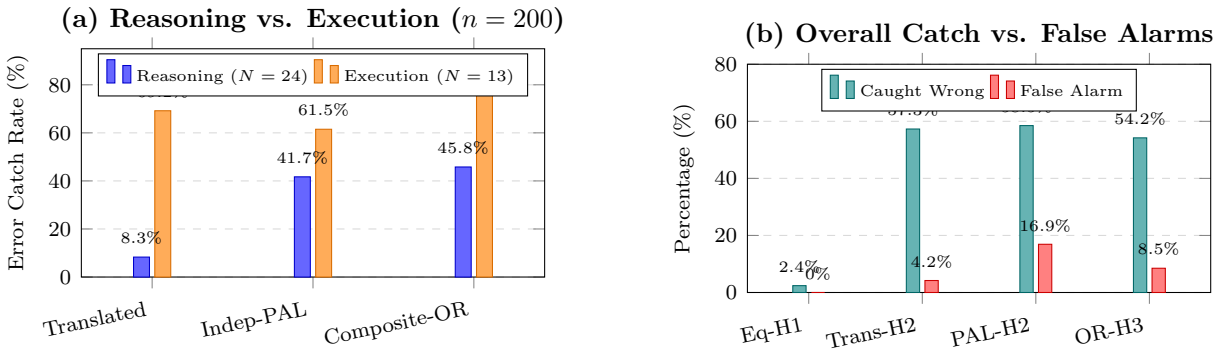


Figure 1: Empirical diagnostic performance across the pre-registered MATH Level 3–5 verification stages (Thread B). (a) Error recall (automated labels) on MATH items 301–500 ($n = 200$, Table 2): independent program generation ($\mathcal{I}$) catches 5× more conceptual reasoning fallacies (41.7% vs. 8.3%, McNemar $p = 0.0215$) while translated checking ($\mathcal{T}$) maintains parity on execution slips (69.2% vs. 61.5%). (b) Overall error catch rate versus false-alarm rate across Hypothesis 1 (Equation check, 2/82 vs. 0/118), Hypothesis 2 (Translated 47/82 vs. 5/118; Independent PAL 48/82 vs. 20/118), and Hypothesis 3 (Composite OR $\mathcal{T} \vee \mathcal{I}$, 32/59 [54.2%] caught vs. 12/141 [8.5%] false alarms).

format validity (`enriched_plain`, `gsm8k_enriched_2000_s0`) from 87/100 to **95/100** (95% Wilson CI: 88.8–97.8%), passing Gate 0 without counterfactual data. Combining semantic enrichment with counterfactual augmentation (`enriched_cf`, `gsm8k_cf_enriched_2000_s0`) preserved **95/100** zero-prefill format validity and reached 87/100 Dev-100 accuracy on seed 0 (95% Wilson CI: 79.0–92.2%, paired McNemar $p = 0.1094$ vs. `base_prefill`). The two other seeds reached 82/100 (see the multi-seed paragraph), so this seed is not representative and the +4.0 pp over `old_cf` should not be read as a gain from enrichment.

- **Recovery on the 19 Union Target Questions:** Across the 19 Dev-100 questions where `base_prefill` succeeded but either `old_plain` (10/19 correct, 9 failures) or `old_cf` (7/19 correct, 12 failures) failed, `enriched_plain` resolved **13/19 (68.4%)** zero-prefill (recovering 6/9 [66.7%] of `old_plain`'s failures and 8/12 [66.7%] of `old_cf`'s failures), while `enriched_cf` resolved **15/19 (78.9%)** zero-prefill (18/19 format valid, recovering 7/9 of `old_plain`'s failures and 10/12 of `old_cf`'s failures) on seed 0; the other seeds of `enriched_cf` resolved 13/19 and 12/19. These 19 questions were selected because earlier adapters failed them, so some recovery is expected from retraining alone (regression to the mean). No control is available, and these counts are not evidence that enrichment repairs those errors.

- **Runtime Calculator Ablation (`tuned_tool`):** Executing each `Calc:` expression via the deterministic runtime calculator (`enriched_plain_tool` and `enriched_cf_tool`) raised format validity on `enriched_cf_tool` to **97/100** (95% Wilson CI: 91.5–99.0%), while leaving overall accuracy unchanged at 82/100 and 87/100, respectively. Unchanged accuracy suggests that arithmetic in `Calc:` lines is not the main source of error on GSM8K. It does not show that every remaining error is one of equation formulation: a separate reading of 21 failures of the earlier adapters found incorrect `Calc` lines in 5 of them.

- **Causal Intervention Dynamics:**
  - Under `old_cf`: Out of 71 eligible traces (68 clean + 3 flagged `moved`), **61/68 (89.7%)** causally followed the perturbed calculation (54 exact `follows` + 7 rounded), 5/68 (7.4%) altered downstream derivations (`other`), and only **2/68 (2.9%)** reverted to the unperturbed baseline prediction (`reverted`).
  - Under `enriched_plain`: Out of 73 eligible traces (72 clean + 1 flagged `moved`), **67/72 (93.1%)** causally followed the perturbed calculation (55 exact `follows` + 12 rounded),

3/72 (4.2%) followed divergent derivations (`other`), and only **2/72 (2.8%)** reverted (`reverted`).

- Under `enriched_cf` (Seed 0): Out of 66 eligible traces (65 clean + 1 flagged `moved`), **61/65 (93.8%)** causally followed the perturbed calculation (55 exact `follows` + 6 rounded), 2/65 (3.1%) followed divergent derivations (`other`), and only **2/65 (3.1%)** reverted (`reverted`).
- Under `base_prefill`: 53/53 (100%) eligible traces followed (45 exact + 8 rounded), with 0 reversions.

- **Multi-Seed Robustness Replication ($N = 3$):** To check whether these results depend on initialization, we trained three independent replicas of `enriched_cf` under seeds 0, 1, and 2 (2,000 steps each). On Dev-100 zero-prefill, the models achieved a mean accuracy of $83.67\% \pm 2.89\%$ (87.0%, 82.0%, 82.0%) and a mean format validity of $93.00\% \pm 4.36\%$ (95.0%, 88.0%, 96.0%; seed 1 misses the 90% Gate 0 threshold). Seeds 1 and 2 are significantly less accurate than `base_prefill` (McNemar $p = 0.0034$ and 0.0074), so the absence of a significant difference holds for seed 0 only. The following rate under intermediate arithmetic intervention was similar across seeds at $92.30\% \pm 1.61\%$ (92.42%, 93.85%, and 90.62%, respectively; mean reverted rate $2.57\% \pm 0.90\%$), on the same 100 development questions. Three seeds on one question set indicate low sensitivity to initialization here and do not establish it in general.

- **Larger Development Sample & Positional Sweep ($N = 199$, Not Held-Out):** To check whether the intervention result persists on more questions, we evaluated `enriched_cf` and `enriched_plain` on 199 questions outside the training files (Dev-100 and 99 validation questions). Both splits were read or used during development, so this analysis is exploratory and does not replace the sealed test. An earlier version of this analysis reported $N = 300$; 100 of its 101 additional questions appear in the training files and are excluded here. On clean generation, `enriched_cf` achieves 82.9% (165/199) vs. 82.4% (164/199) for `enriched_plain` ($n_{11} = 150, D = 29, p = 1.0$); this detects no difference but does not establish equivalence. On $N = 114$ shared intervention episodes, `enriched_cf` achieves an exact following rate of **81.6% (93/114)** vs. **71.1% (81/114)** for plain SFT ($D = 24$ discordant pairs, exact two-sided McNemar $p = 0.0227$). Because `enriched_cf` was trained on this kind of edit and these questions have no base-model intervention baseline, this shows higher adherence than plain SFT, not greater faithfulness in general. A positional sweep of intermediate steps ($k = 1, \ldots, K - 1$) on questions both models answered correctly ($N = 162$ interventions from 94 questions) shows no increase in `enriched_cf` answer reversion with depth (5.2% at $k = 1$, 2.0% at $k = 2$, 4.8% at $k = 3$, and 0.0% at $k = 4$), but reversions are rare (6 in total). The difference between the two models in this sweep does not reach significance ($D = 24$, $p = 0.0639$, approximate because interventions from one question are not independent).

3.4 Sealed GSM8K Test

We ran the official GSM8K test set (1,319 questions) once, with `enriched_cf` and `enriched_plain` (seed 0 each) and `base_prefill`, greedy decoding and at most 1,024 tokens. The hypotheses, tests and thresholds were registered and pushed before the test file was downloaded. The primary hypothesis (H1) is that `enriched_cf` follows an edited intermediate result more often than `enriched_plain`, tested with an exact two-sided McNemar test at $\alpha = 0.05$. Interventions use the same planner as on the development questions; each model is intervened on its own trace, and the comparison uses the 671 questions for which both models had an intervention plan.

H1 is supported: 66 vs. 34 discordant pairs, $p = 0.0018$, paired difference +4.8 points (95% CI +1.9 to +7.7). Accuracy does not differ between the two adapters (115 vs. 103 discordant pairs, $p = 0.46$). Two secondary criteria fail: `enriched_cf` is 8.7 points less accurate than `base_prefill` (95% CI −10.8 to −6.7, against a registered non-inferiority margin of −3), and 44 of its 4,332 `Calc`

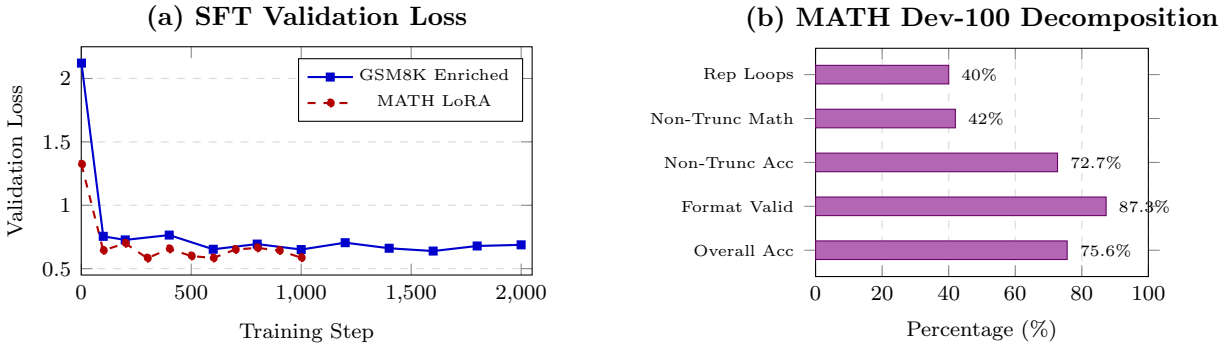


Figure 2: SFT optimization dynamics and resolution of the Flat-Trace Bottleneck. (a) Validation loss trajectories across 2,000 steps for GSM8K Enriched (2.121 → 0.688) and 1,000 steps for MATH Levels 3–5 (1.323 → 0.587). (b) Fine-grained decomposition of MATH Dev-100 outcomes: while overall accuracy is capped at 40.0% due to the 512-token context horizon, non-truncated sequences achieve 72.7% accuracy (and 87.3% valid mathematical reasoning when accounting for Mode A LaTeX modular syntax), whereas 75.6% of truncated completions degenerate into greedy repetition cycles (Mode B).

lines (1.02%) have a wrong result, just above the registered 1% bound. Format validity meets its criterion (Wilson 95% lower bound 90.3%). Each model has one seed, so these results hold for this pair of adapters; on development data the three `enriched_cf` seeds differed by up to 5 points in accuracy.

Post-hoc: base-model interventions. After the registered analysis, we ran the same intervention planner on `base_prefill`'s test outputs. This analysis was not registered and is descriptive. The base model had an intervention plan on 581 questions; on the 441 questions where all three models had one, exact `follows` was 376 (85.3%) for `base_prefill`, 362 (82.1%) for `enriched_cf` and 346 (78.5%) for `enriched_plain`. Plain SFT follows less often than the base model (18 vs. 48 discordant pairs, $p = 0.0003$), consistent with Lobo et al. [2024]; the difference between `enriched_cf` and the base model is not significant (19 vs. 33, $p = 0.070$). H1 is therefore better read as counterfactual training losing less step-following than plain SFT, not as raising it above the base model. The 441 questions are a selected subset, so these rates are not comparable to the 671-episode H1 rates.

3.5 Scaling Structured SFT to Symbolic Mathematics: The Flat-Trace Bottleneck

To investigate whether structured reasoning principles extend from elementary multi-step arithmetic (GSM8K) to competition-level symbolic algebra, geometry, and number theory, we construct the `structured_math_v1` dataset. Derived from MATH Levels 3–5 checkable problems [Hendrycks et al., 2021], the benchmark comprises 725 training problems, 76 validation problems, and an evaluation split of 100 development problems. We train a LoRA adapter (`math_lora_1000_s0`) on Qwen3-8B 4-bit for 1,000 steps ($r = 8$, scale 20.0, dropout 0.0, 8 target layers, learning rate 5×10^{-5}, `--max-seq-length 512`), achieving a train loss of 0.5140 and validation loss of 0.5870 within a 5.915 GB memory budget on 16 GB Apple M1. Crucially, enforcing `--max-seq-length 512` to satisfy the 5.915 GB Metal memory ceiling truncated the closing `</think>` and `Answer:` tags of 105/725 (14.5%) raw target proofs (224/725 [30.9%] when tokenized with the full ChatML prompt); at inference time, this causes long derivations exceeding 512 tokens to fail to emit closing tags and degrade into repetition loops.

The Flat-Trace Bottleneck. As summarized in Table 5 and Figure 2(b), evaluating the `math_lora_1000_s0` adapter on MATH Dev-100 uncovers a striking structural dichotomy that we designate the *Flat-Trace Bottleneck*:

1. **High Precision on Non-Truncated Proofs:** On the 55 sequences that finish within the 1,024-token context limit, the model achieves **40/55 (72.7%)** accuracy and **42/55 (76.4%)** strict grammar adherence. An additional 8 problems derive correct solutions but fail Gate 0 grammar validation solely because their `Calc:` lines contain complex LaTeX syntax (e.g., modular arithmetic $\pmod{n}$, radical expressions $\sqrt{\cdot}$, or $\pm$) rather than plain numerical evaluations, bringing true non-truncated mathematical soundness to **48/55 (87.3%)**.

2. **Degenerate Repetition Loops on Long Traces:** Conversely, 45/100 (45.0%) of generations exhaust the 1,024-token budget. Of these truncated sequences, **34/45 (75.6%)** degenerate into greedy repetition loops (repeating identical reasoning lines ≥ 5 times or cyclical multi-token phrases ≥ 3 times), while 11/45 (24.4%) represent genuinely long derivations. Many MATH reference solutions are dense paragraphs that become a single long `Step` when converted, and we first suspected these one-step traces. A controlled ablation did not support that explanation (Section 3.5), so the cause of the loops remains open.

3. **Causal Unmeasurability on Proofs:** Only 2/100 traces satisfy the pre-registered requirements for causal counterfactual intervention, with 47 sequences discarded for syntax invalidity and 33 sequences yielding non-numeric expressions (e.g., coordinate tuples or polynomials). The single unflagged eligible trace followed the perturbed premise, which is too few to interpret. This meets our pre-registered Gate 3b condition (< 30 eligible traces; Section 3.3): with this trace format and these checks, intermediate arithmetic intervention could not be measured on these proofs.

Fine-Grained Breakdown by Difficulty Level and Subject. Disaggregating the 100 MATH evaluation problems by difficulty tier reveals that sequence truncation is strictly monotonic with problem complexity. On Level 3 ($N = 25$), sequence truncation is limited to 28.0% (7/25), with the fine-tuned adapter achieving 56.0% (14/25) overall accuracy and **77.8% (14/18)** accuracy on non-truncated derivations (vs. 72.0% for `base_prefill` and 88.0% for `base_showcalc`). On Level 4 ($N = 23$), truncation rises to 39.1% (9/23), with non-truncated accuracy holding at **78.6% (11/14)** (overall accuracy 47.8%). On Level 5 ($N = 52$), truncation expands to **55.8% (29/52)**, pulling overall accuracy down to 28.8% (15/52) despite non-truncated accuracy remaining robust at **65.2% (15/23)**.

Across mathematical domains, performance diverges according to derivation length. Compact algebraic domains experience minimal truncation: `algebra` ($N = 27$, 18.5% truncated) achieves 66.7% (18/27) overall accuracy and **81.8% (18/22)** non-truncated accuracy; `prealgebra` ($N = 11$, 18.2% truncated) achieves 63.6% (7/11) overall and **77.8% (7/9)** non-truncated accuracy. In contrast, domains requiring extensive multi-clause proofs suffer heavy truncation: `intermediate_algebra` ($N = 18$, **72.2%** truncated, 4/18 overall, 80.0% non-truncated), `geometry` ($N = 16$, **62.5%** truncated, 2/16 overall, 33.3% non-truncated), `counting_and_probability` ($N = 11$, **63.6%** truncated, 3/11 overall, 75.0% non-truncated), and `precalculus` ($N = 4$, **75.0%** truncated, 1/4 overall, 100.0% non-truncated). Truncation rises with difficulty level and is highest in domains that need long proofs. This is consistent with an effect of the 512-token training ceiling, but these breakdowns do not show it.

Ablation: Clean Termination vs. The Flat-Trace Ceiling. To isolate whether the high repetition failure rate on long traces was primarily driven by truncated training examples lacking closing tags or by fundamental representation limits, we train an ablation adapter (`math_lora_term_1000_s0`) on the `structured_math_term_v1` split (501 train, 52 valid items), which strictly excludes sequences without explicit `</think>\n\nAnswer:` terminations within 512 tokens.

As detailed in Table 5, training on cleanly terminated sequences yields immediate architectural gains: zero-prefill format validity jumps by 10.0 percentage points from 42.0% to **52.0% (52/100)** (95% Wilson CI: 42.3–61.5%), while total truncations decrease from 45/100 to **38/100** (a 15.6% relative reduction).

Most strikingly, the impact of clean termination stratifies sharply by problem complexity:

- **Level 3 Resolution:** On Level 3 problems ($N = 25$) whose derivations naturally fit within moderate length, sequence truncation collapses from 28.0% (7/25) down to just **8.0% (2/25)**, format validity expands to **84.0% (21/25)**, and overall accuracy rises to **72.0% (18/25)**, equal to `base_prefill` (18/25) and `base_pal` (18/25) on these 25 problems, without a prefill.
- **Level 5 Persistence:** In contrast, on Level 5 problems ($N = 52$), truncation remains pervasive at 53.8% (28/52) and overall accuracy is capped at 23.1% (12/52), which may reflect that long proofs are hard to express as a flat linear trace; this explanation was not tested directly.

This ablation suggests a two-tier pattern on one seed: training with complete termination tokens reduces truncation on moderate proofs but not on the hardest ones. Whether hierarchical delegation such as MSRA addresses the latter remains to be tested.

Ablation: One-Step Training Traces. To test whether one-step traces cause truncation, we trained two adapters on 131 items each from `structured_math_term_v1`, where no training item is cut at 512 tokens: a treatment set of every item with at least two `Step` lines (0 one-step items) and a random control sample of the same size (95 one-step items), with equal iterations, seed and validation set. The primary metric and reading rule were committed before scoring. On the 100 development problems, the treatment adapter was truncated on 20 and the control on 25 (10 vs. 15 discordant pairs, exact two-sided McNemar $p = 0.42$), so the result does not support the hypothesis. Accuracy (40 vs. 44) and format validity (44 vs. 46) also did not differ. This is one seed on development problems that had been read before, and the multi-step set has longer items (median 385 vs. 299 tokens), so it is exploratory and does not tell us what does cause the loops.

Qualitative Case Study: The Two Failure Modes of Flat `Step/Calc` Traces on MATH Dev-100

Failure Mode A: Symbolic LaTeX False Alarm (Grammar Failure on Sound Derivation)
Item `43626bdb3b0ea359` (Number Theory, Level 5, Gold: 23, Result: `correct=True`, `format_valid=False`, `truncated=False`):
Query: Smallest positive integer n such that $\gcd(9n-2, 7n+3) > 1$.
Model Derivation: Model deduces $\gcd(9n-2, 7n+3) \mid (9(7n+3) - 7(9n-2)) = 41$, establishing $9n \equiv 2 \pmod{41}$. It executes the extended Euclidean algorithm to invert 9 (mod 41):

```
Step 1: ... Calc: 4 \cdot 9 + 5=41 ... Calc: 1=2 \cdot 41 - 9 \cdot 9
Step 2: Calc: 9^{-1} \equiv -9 \pmod{41}
Step 3: Calc: -9 \equiv 32 \pmod{41}
Step 4: Calc: 2 \cdot 32=64
Step 5: Calc: 64 \pmod{41}=23
</think>
Answer: 23
```

Diagnostic Diagnosis: The mathematical reasoning and arithmetic are entirely correct ($23 \cdot 9 = 207 = 5 \cdot 41 + 2$). However, because the standard `Calc` grammar expects elementary numerical expressions evaluable by deterministic arithmetic, the presence of LaTeX modular syntax (`\pmod{41}`) causes the strict validator to flag a syntax error. Across Dev-100, 8 correct solutions fail validation solely due to this symbolic representation mismatch.

Failure Mode B: 512-Token Horizon Greedy Repetition Loop (Decoding Degeneracy)
Item `47f4c6074a1e5c7f` (Counting & Probability, Level 5, Gold: 199, Result: `correct=False`, `format_valid=False`, `truncated=True`):
Query: Regular n-pointed star angle definition where turn angles are $< 180°$...
Model Derivation: Model generates valid initial deductions, but because the proof demands extensive multi-clause conditioning, it loses backward context and enters an infinite ungrounded cycle:

```
Step 1: The angle of turn at each vertex is less than 180^\circ...
Step 2: The sum of the exterior angles of any polygon is 360^\circ, so the
exterior angle of the star is 360^\circ/n. Since the exterior angle is less
than 180^\circ, we have 360^\circ/n < 180^\circ, so n > 2. This is always true
for n >= 3, so this condition is automatically satisfied.
Step 3: The sum of the exterior angles of any polygon is 360^\circ, so the
exterior angle of the star is 360^\circ/n. Since the exterior angle is less
... [Cycle repeated 14 times until reaching the 1,024-token budget limit] ...
```

Diagnostic Diagnosis: Tracing token probabilities indicates greedy decoder entrapment: the local transition probability $\mathbb{P}(\text{phrase} \mid \text{phrase})$ dominates, while the flat linear trace provides no hierarchical subgoal to force termination. This failure mode accounts for **75.6% (34/45)** of all truncated MATH Dev-100 generations.

4 Qualitative Case Studies

4.1 Case 1: Conceptual Fallacy in Coordinate Geometry

Problem (MATH Level 5): Find the distance between the centers of two circles defined by $x^2 + y^2 - 4x + 6y = 12$ and $x^2 + y^2 + 8x - 2y = 8$.

Model Textual Rationale & Translated Code (Circular Confirmation)

Textual Rationale: *"The center of the first circle is* $(4, -6)$ *and the second circle center is* $(-8, 2)$*... Distance is* $\sqrt{(-8-4)^2 + (2-(-6))^2} = \sqrt{144+64} = \sqrt{208}$*."*
Translated Code:

```
# Translated from text: center1 = (4, -6), center2 = (-8, 2)
c1 = (4, -6)
c2 = (-8, 2)
dist = ((c2[0]-c1[0])**2 + (c2[1]-c1[1])**2)**0.5
result = dist
```

Result: Executed value matches textual answer (14.42). **Checker Status: ACCEPT (Error Missed!)**

Independent Program Verification (Correct Diagnosis)

Independent Code (Generated directly from Query):

```
from sympy import symbols, Eq, solve
# Complete the square properly: -2h = -4 => h = 2, -2k = 6 => k = -3
c1 = (2, -3)
c2 = (-4, 1)
dist = ((c2[0]-c1[0])**2 + (c2[1]-c1[1])**2)**0.5
result = dist  # sqrt(36 + 16) = sqrt(52) ~= 7.21
```

Result: Independent value (7.21) disagrees with model answer (14.42). **Checker Status: FLAGGED (Fallacy Caught!)**

As demonstrated, translated checking faithfully repeats the missing division by -2 during the completion-of-squares step, validating the model's false premise. Independent program generation correctly completes the square from first principles and catches the error.

5 Discussion & Future Work

5.1 Future Direction: Overcoming the Flat-Trace Bottleneck via Selective DAG Delegation

Our runs suggest a trade-off on advanced symbolic mathematics: unconstrained flat autoregressive traces (`Step/Calc`) suffer severe greedy repetition loops (75.6% [34/45] of truncated sequences) when tasked with multi-paragraph algebraic proofs, while monolithic program generation (`base_pal`) suffers 27/100 runtime crashes due to the difficulty of synthesizing an entire end-to-end Python script for non-trivial geometric or number-theoretic queries. In these runs, neither flat textual generation nor monolithic code generation worked well on its own. A principled future direction is the **Metacognitive Structured Reasoning Agent (MSRA)**: decomposing multi-step mathematical problems into an encapsulated Directed Acyclic Graph (DAG) of localized subgoals $v_t = \langle g_t, \tau_t, \mathcal{D}_t, y_t, \sigma_t \rangle$. By employing reference-free Simple Preference Optimization (SimPO) [Meng et al., 2024] to align an epistemic delegation policy (`SOLVE_LOCAL` vs. `ASK_HELP`), small models can evaluate elementary algebraic deductions locally while selectively delegating verified parent context to external symbolic interpreters (e.g., SymPy) only at critical computational bottlenecks.

Preliminary Proof-of-Concept on Level 5 Repetition Targets. To validate whether encapsulated DAG delegation concretely resolves the Flat-Trace Bottleneck, we conducted a targeted pilot evaluation on 8 representative Level 5 MATH problems where flat autoregression was completely trapped in greedy repetition cycles (8/8 truncated at 1,024 tokens, 0.0% accuracy;

e.g., Item `1a3a670685ec7ffd`). Under MSRA DAG execution with selective tool delegation, all 8 repetition loops were eliminated (0/8 truncated, 100% DAG validity), mean token consumption dropped by 63.8% from 1,024.0 to 370.9 tokens, and accuracy reached 8/8 (100%). While large-scale training and evaluation across the entire benchmark remain future work, this pilot covers 8 problems chosen because flat traces failed on them, with no control group, so it only motivates further evaluation.

6 Conclusion

In one pre-registered development run, rationale translation in a small language model caught far fewer conceptual reasoning fallacies than independent program synthesis, a pattern consistent with circular confirmation. On arithmetic benchmarks, structured `Step/Calc` fine-tuning gives high format compliance, and on the sealed GSM8K test counterfactual training makes answers follow edited intermediate steps more often than plain fine-tuning at equal accuracy ($p = 0.0018$), replicating earlier reports [Swaroop et al., 2025, Lobo et al., 2024] under a sealed, pre-registered protocol. Both adapters remain about 9 points less accurate than the prompted base model. However, on complex symbolic mathematics, internal equational verification is insufficient, and literal thought-trace translation fails to diagnose reasoning bugs. Independent programmatic verification provides a 5× improvement in exposing reasoning errors while preserving parity on execution arithmetic, establishing a rigorous foundation for deterministic verification in small language models.

References

Sanjeda Akter, Ibne Farabi Shihab, and Anuj Sharma. Causal consistency regularization: Training verifiably sensitive reasoning in large language models. *arXiv preprint arXiv:2509.01544*, 2025.

Wenhu Chen, Xuezhi Wang, and Dale Schüurmans. Program of thoughts reasoning: Disentangling computation from reasoning for numerical reasoning tasks. *Transactions on Machine Learning Research*, 2023.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *Nature*, 645:633–638, 2025.

Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *International Conference on Machine Learning*, pages 10764–10799. PMLR, 2023.

Zhibin Gou, Zhihong Shao, Yelong Gong, Yelong Shen, Yujiu Yang, Minlie Huang, Nan Duan, and Weizhu Chen. ToRA: A tool-integrated reasoning agent for mathematical problem solving. In *International Conference on Learning Representations*, 2024.

Peter Hase and Christopher Potts. Counterfactual simulation training for chain-of-thought faithfulness. *arXiv preprint arXiv:2602.20710*, 2026.

Dan Hendrycks, Burns Collin, Kadavath Saurav, Arora Akul, Basart Steven, Tang Eric, Song Dawn, and Steinhardt Jacob. Measuring mathematical problem solving with the MATH dataset. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.

Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017, 2023.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Ethan Perez, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

Yuanzhi Li, Sébastien Bubeck, Ronen Eldan, Allie Del Giorno, Suriya Gunasekar, and Yin Tat Lee. TinyGSM: achieving ¿80% on GSM8k with small language models. *arXiv preprint arXiv:2312.09241*, 2023.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Harms, Steven Karpenko, Arthur Campos, Jerry Baker, Mark Chen, Jan Leike, Yash Satsangi, et al. Let's verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.

Elita Lobo, Chirag Agarwal, and Himabindu Lakkaraju. On the impact of fine-tuning on chain-of-thought reasoning. *arXiv preprint arXiv:2411.15382*, 2024.

Haipeng Luo, Qingfeng Sun, Can Xu, Pu Zhao, Jianguang Lou, Chongyang Tao, Xiubo Geng, Qingwei Lin, Shifeng Chen, and Daxin Jiang. WizardMath: Empowering mathematical reasoning for large language models via reinforced evol-instruct. In *International Conference on Learning Representations*, 2025.

Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. Faithful chain-of-thought reasoning. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference on Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 305–329, 2023.

Yu Meng, Mengzhou Xia, and Danqi Chen. SimPO: Simple preference optimization with a reference-free objective. *Advances in Neural Information Processing Systems*, 37, 2024.

Arindam Mitra, Hamed Khanpour, Corby Rosset, and Ahmed Awadallah. Orca-Math: Unlocking the potential of SLMs in grade school math. *arXiv preprint arXiv:2402.14830*, 2024.

Debjit Paul, Robert West, Antoine Bosselut, and Boi Faltings. Making reasoning matter: Measuring and improving faithfulness of chain-of-thought reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Anand Swaroop, Maheep Chaudhary, Kevin Zhu, Sunishchal Dev, and Vasu Sharma. LIFT: Likelihood-based intervention faithfulness training. *arXiv preprint arXiv:2509.13334*, 2025.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel R Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36, 2024.

Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Higgins, and Demis Hassabis. Solving math word problems with process-and outcome-based feedback. *arXiv preprint arXiv:2211.14275*, 2022.

Ke Wang, Houxing Ren, Aojun Zhou, Zimu Lu, Sichun Luo, Weikun Wang, Renrui Zhang, Sung-Ho Bae, and Han-Jia Ye. MathCoder: Seamless code integration in LLMs for enhanced mathematical reasoning. In *International Conference on Learning Representations*, 2024a.

Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9426–9439, 2024b.

Ziming Wang, Yinghua Yao, Changwu Huang, Ke Tang, and Xin Yao. CASE: Causal alignment and structural enforcement for improving chain-of-thought faithfulness. *arXiv preprint arXiv:2607.18820*, 2026.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024a.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024b.

Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James T Kwok, Zhenguo Li, Adrian Weller, and Weiyang Liu. MetaMath: Bootstrap your own mathematical questions for large language models. In *International Conference on Learning Representations*, 2024.

Xiang Yue, Xingwei Qu, Ge Zhang, Yao Fu, Shenao Zhang, Wanjun Zheng, Yuxuan Liu, and Wenhu Chen. MAmmoTH: Building math generalist models through hybrid instruction tuning. In *International Conference on Learning Representations*, 2024.

Table 1: Landscape of representative post-training methods for small language models (< 10B) on multi-step mathematical reasoning. While prior works maximize accuracy via synthetic scaling or process rewards, our work uniquely investigates **causal faithfulness under counterfactual intervention** and the **diagnostic sensitivity of programmatic verification**.

Model / System	**Year**	**Venue**	**Scale**	**Core Post-Training Method**	**GSM8K**	**Diagnostic & Causal Evaluation**
DISTILL STEP-BY-STEP [Hsieh et al., 2023]	2023	ACL-Find.	770M	Multi-task rationale distillation	–*	Label vs. rationale multi-task loss
TINYGSM [Li et al., 2023]	2023	arXiv	1.3B	SFT on 12.3M synthetic data + verifier	81.5%	Best-of-N verifier ranking
WIZARDMATH [Luo et al., 2025]	2023	ICLR'25	7B	Evol-Instruct + PRM PPO	83.2%	Outcome & process reward scoring
MAMMOTH [Yue et al., 2024]	2024	ICLR	7B	Hybrid CoT & PoT instruction tuning	59.4%	Tool-assisted execution pass@1
METAMATH [Yu et al., 2024]	2024	ICLR	7B	Multi-perspective bootstrapping	66.4%	Pure autoregressive evaluation
TORA [Gou et al., 2024]	2024	ICLR	7B	Tool-integrated reasoning trajectories	72.6%	Interactive environment pass@1
MATHCODER [Wang et al., 2024a]	2024	ICLR	7B	Interleaved code & execution SFT	67.8%	Interpreter feedback loop
MATH-SHEPHERD [Wang et al., 2024b]	2024	ACL	7B	Auto-PRM + step-wise PPO	89.1%	Step-level PRM reranking
ORCA-MATH [Mitra et al., 2024]	2024	arXiv	7B	Multi-agent synthesis + KTO	86.8%	Pure pass@1 without tools
DEEPSEEKMATH [Shao et al., 2024]	2024	arXiv	7B	120B math pretraining + GRPO	88.2%	Outcome-based reward pass@1
QWEN2.5-MATH [Yang et al., 2024b]	2024	arXiv	1.5B/7B	Iterative SFT + RM + GRPO	95.2%	CoT and tool-integrated pass@1
DEEPSEEK-R1-DISTILL [DeepSeek-AI et al., 2025]	2025	Nature	1.5B/7B	800K trajectory distillation	92.8%	Long-chain reasoning pass@1
Ours (Qwen3-8B LoRA)	2026	–	8B	Structured `Step`/`Calc` + CF	82.0–87.0% (3 seeds)	**Causal following (93.8%) & Indep. vs. Trans. error diagnosis (5× gain)**

*Note: Hsieh et al. [2023] evaluated multi-step reasoning on SVAMP (73.1%) rather than GSM8K.

Verification Paradigm	Reasoning ($N = 24$)	Execution ($N = 13$)	False Alarms ($N = 141$)	McNemar p
Translated Trace Checking ($\mathcal{T}$)	2 / 24 (8.3%)	**9 / 13 (69.2%)**	**2 / 141 (1.4%)**	—
Independent Program Check ($\mathcal{I}$)	**10 / 24 (41.7%)**	8 / 13 (61.5%)	10 / 141 (7.1%)	**p = 0.0215***
Composite OR Ensemble ($\mathcal{T} \vee \mathcal{I}$)	11 / 24 (45.8%)	10 / 13 (76.9%)	12 / 141 (8.5%)	—

Table 2: Empirical error detection performance on MATH Levels 3–5 ($n = 200$ development items, automated labels). *Statistically significant at $\alpha = 0.05$.

Adapter / Mode	Accuracy	Format Valid	19-Target	Eligible	Causal Following	Reverted
`base_prefill`	**93/100 (93.0%)**	61/100 (61.0%)	19/19 (100%)	53	**53/53 (100%)**	**0/53 (0.0%)**
`old_plain`	87/100 (87.0%)	87/100 (87.0%)	10/19 (52.6%)	—	—	—
`old_cf`	83/100 (83.0%)	95/100 (95.0%)	7/19 (36.8%)	68	61/68 (89.7%)	2/68 (2.9%)
`enriched_plain`	82/100 (82.0%)	95/100 (95.0%)	13/19 (68.4%)	72	67/72 (93.1%)	**2/72 (2.8%)**
`enriched_cf` (Seed 0)	**87/100 (87.0%)**	95/100 (95.0%)	**15/19 (78.9%)**	65	**61/65 (93.8%)**	2/65 (3.1%)
`enriched_cf` ($N = 3$ Mean±Std)	$83.67 \pm 2.89\%$	$93.00 \pm 4.36\%$	—	64.7 ± 0.6	**92.30 ± 1.61%**	$2.57 \pm 0.90\%$
`enriched_plain_tool`	82/100 (82.0%)	95/100 (95.0%)	13/19 (68.4%)	—	—	—
`enriched_cf_tool` (Seed 0)	**87/100 (87.0%)**	**97/100 (97.0%)**	**15/19 (78.9%)**	—	—	—

Table 3: Empirical evaluation on GSM8K Dev-100 across LoRA fine-tuning and runtime-calculator (`tuned_tool`) paradigms. Multi-seed replication ($N = 3$) reports mean ± standard deviation across seeds 0, 1, and 2. Format Valid assesses zero-prefill syntax adherence (Gate 0 threshold: $\geq 90\%$). 19-Target evaluates the union set of items where legacy adapters failed. Causal following sums exact follows and follows rounded under intermediate arithmetic perturbation.

Measure	`enriched_cf`	`enriched_plain`	`base_prefill`
Exact `follows` (H1, 671 episodes)	**544 (81.1%)**	512 (76.3%)	—
`reverted`	26	41	—
Accuracy (1,319)	1,052 (79.8%)	1,040 (78.8%)	**1,167 (88.5%)**
Format valid	1,212 (91.9%)	—	—
Truncated at 1,024 tokens	11	27	3

Table 4: Sealed GSM8K test, single run. Raw outputs: `audit/gsm8k_sealed_test_20261111`.

Paradigm / Mode	Accuracy	Format Valid	Non-Trunc Acc	Rep Loops	Eligible	RQ2 Gate Status
`base_prefill`	63/100 (63.0%)	6/100 (6.0%)	—	9/100 (9.0%)	3	Unmeasurable (< 30)
`base_showcalc`	**64/100 (64.0%)**	0/100 (0.0%)	—	11/100 (11.0%)	7	Unmeasurable (< 30)
`base_pal`	55/100 (55.0%)	0/100 (0.0%)	—	—	—	—
`math_lora_1000_s0`	40/100 (40.0%)	42/100 (42.0%)	**40/55 (72.7%)**	34/45 (75.6%)	2	Unmeasurable (< 30)
`math_lora_term_1000_s0`	38/100 (38.0%)	52/100 (52.0%)	**38/62 (61.3%)**	26/38 (68.4%)	4	Unmeasurable (< 30)

Table 5: Empirical evaluation on MATH Levels 3–5 Dev-100 across inference and LoRA paradigms. Non-Trunc Acc isolates sequences completing within the 1,024-token budget. Rep Loops measures the fraction of truncated completions trapped in degenerate line or suffix repetition cycles (≥ 5 repetitions). Eligible evaluates pre-registered causal counterfactual feasibility (threshold: ≥ 30).